Dr. Adrian Gildardo Gutierrez Vazquez

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Profile

Experimental particle physicist with over 9 years of research experience in high-energy physics and a strong background in computational methods and data analysis. Lead analyst for the electromagnetic calorimeter (ECAL) in the MUonE experiment at CERN. Expertise in machine learning, statistical modeling, detector reconstruction, and large-scale data analysis in high-noise environments.

Education

University of Virginia — Charlottesville, VA
Ph.D. in Particle Physics — 2026
Advisor: Dinko Počanić

University of Virginia — Charlottesville, VA
M.S. in High Energy Physics — 2022
Advisor: Robert Hirosky
Thesis: Simulation of Magnetic Monopoles in 14 TeV Proton-Proton Collisions at the CMS Electromagnetic Calorimeter

University of Oregon — Eugene, OR
B.S. in Physics (Honors) & Mathematics — 2019
Advisor: Stephanie Majewski
Thesis: Edge Detection and Deep Learning Algorithm Performance Studies for the ATLAS Trigger System

Mount Hood Community College — Gresham, OR
Associate in Science — 2015

Research Experience

MUonE Experiment, CERN — 2022 – 2026
Graduate Researcher / Lead ECAL Analyst

- Led the analysis of the electromagnetic calorimeter (ECAL) for precision muon–electron scattering measurements
- Developed reconstruction algorithms for calorimeter signals, including gain calibration, pulse reconstruction, and tracker–calorimeter matching
- Designed and implemented data processing pipelines using Python, C++, and ROOT
- Led integration of the ECAL into the experiment's data acquisition (DAQ) system, defining data reconstruction architecture and storage strategy
- Performed detector performance studies using both experimental data and GEANT4-based simulations (FairMUonE)
- Mentored junior graduate students and coordinated analysis tasks within the collaboration

CMS Experiment — 2019 – 2022
Graduate Researcher

- Simulated magnetic monopole and photon production using GEANT4 within the CMS detector framework
- Developed pulse-shape reconstruction techniques for signal-background discrimination
- Studied monopole trajectories through electromagnetic fields using Maxwell tensor formalism
- Contributed to trigger development strategies for rare signal detection
- Applied machine learning techniques using TensorFlow, TMVA, and Scikit-Learn

ATLAS Experiment 2017 – 2019
Undergraduate Researcher

- Developed Sobel filtering algorithms for particle detection and background rejection
- Performed validation of topoclustering algorithms for the ATLAS trigger system
- Conducted deep learning studies for classification tasks using TensorFlow

Duke University REU Summer 2018
Research Intern

- Developed machine learning models (XGBoost, TMVA, Scikit-Learn) for supersymmetry searches

University of Oregon – Advanced Projects Laboratory 2018

- Recreated spontaneous parametric down-conversion process producing entangled photon pairs

University of Oregon – Optics Research Group 2016 – 2017

- Studied temperature dependence of optical mirror refraction properties

Teaching Experience

University of Virginia
Teaching Assistant 2019 – 2021

- Led undergraduate discussion sections and laboratory sessions in physics
- Provided instruction and mentoring in mathematics and physics courses

Mount Hood Community College
Tutor 2013 – 2015

- Provided academic support in mathematics, chemistry, and physics

Selected Presentations

EPS-HEP Conference	2023
SESAPS Annual Meeting	2024, 2025
CMS Data Analysis School, Fermilab	2020
University of Oregon Research Presentations	2017 – 2019

Awards

APS Outstanding Presentation Award	2019
Duke University REU	2018
Oregon Scientist II Scholarship	2016 – 2018

Technical Skills

Programming: Python, C++, ROOT
Software: LaTeX, Mathematica, MATLAB
Machine Learning: TensorFlow, Scikit-Learn, TMVA
Tools: NumPy, SciPy, Pandas, Linux

Languages

Spanish (native)
English (fluent)